

Vignesh P A

Mobile: +91 6383607045

Email: vigneshpa30@gmail.com

LinkedIn: linkedin.com/in/vigneshpa

Github: github.com/VigneshPA-30

Executive Profile Summary

Results-driven Generative AI Engineer with expertise in designing and deploying enterprise-scale AI solutions leveraging **Large Language Models (LLMs)**, Retrieval-Augmented Generation (**RAG**), **Agentic AI** and Intelligent Automation. Proven experience building **semantic search** platforms, **vector-based** knowledge systems, document intelligence applications and **AI-powered diagnostics solutions** utilizing **LangChain**, **FAISS**, **Pinecone**, **Azure OpenAI**, **OpenAI**, **Gemini** and **Mistral models**. Demonstrates strong capabilities in transforming **unstructured data** into actionable intelligence.

Technical Skills

- **Languages & Frameworks:** Python, C++, PyTorch, LangChain, LangGraph, FastAPI
- **Generative AI & LLMs:** Agentic Workflows, Fine-Tuning (Q-LoRA/PEFT), Transformers, OpenAI, Azure OpenAI, Gemini, Mistral
- **Databases, Tools & Cloud:** Vector Databases (FAISS, Pinecone), Docker, GitHub Actions, Jenkins, Continuous Integration (CI), PostgreSQL, Redis

Work Experience

Gen AI Engineer / Software Engineer | Embitel Technologies (Volkswagen Group) | Bengaluru | Jun 2023 – Present

- Architected a **Generative AI diagnostics platform** using Python and Azure OpenAI to automate the analysis of 650+ complex hardware failure scenarios in the daily CI/CD Pipeline.
- Designed a **Retrieval-Augmented Generation (RAG)** system to process and query 250+ historical incident logs utilizing Large Language Models (LLMs).
- Accelerated system debugging and root-cause analysis by 60% by replacing manual log reviews with intelligent, **context-aware semantic search** capabilities.
- Engineered **CI/CD pipelines and DevOps** workflows utilizing **Jenkins and GitHub Actions** to enforce automated static code analysis and strict quality gates.
- Drove an 87.5% reduction in code vulnerabilities (from 40,000 down to 5,000) through rigorous code quality enforcement, earning a Spot Award for outstanding technical contribution.
- Developed Python-based **data engineering and automation** frameworks to eliminate manual validation efforts and accelerate the software testing cycle.
- Built advanced **data analytics and visualization solutions** leveraging Pandas and Matplotlib to streamline test execution tracking and operational reporting.

Projects Experience

Document Intelligence & RAG Chatbot:

Developed a GEN AI-powered document intelligence platform that processed 85+ research papers and enabled context-aware semantic search using LLMs. Integrated **Pinecone** and **FAISS** within a **Retrieval-Augmented Generation (RAG)** architecture to support 300+ conversational queries with accurate retrieval and contextual memory.

Agentic Workflow Automation:

Developed multi-agent **AI system for automated script generation** & topic research, generating 180+ scripts and reducing content creation time by 80%. Integrated LLMs through FastAPI-based REST services & Dockerized deployments, enabling output generation using **Gemini API**.

Open-Source LLM Fine-Tuning:

Fine-tuned a Mistral-3B model using Q-LoRA and PEFT across 3,500 domain-specific enterprise test cases to **enhance model performance**. Deployed the optimized model in a remote environment, achieving a 70% compilation success rate for AI-generated unit tests.

Education

- Bachelor of Engineering (B.E.) – Electronics & Communication Engineering | Kongu Engineering College | 2023